Live Log over MAVLink

FC Tuning Studio - Flutter developer brief

Firmware pushes the full ST-LINK status line to the G20/phone app via chunked STATUSTEXT. Use together with Phase B (HEARTBEAT, ATTITUDE, PARAM).

1. What the app should show

Live Log = same dense ASCII line as Serial Monitor, about every 1-2 seconds. After reassembly the line looks like:

```
[DMA] s13364 DIS G -0.00 -0.00 -0.00 dps | R -0.60 P 7.70 Y 0.01 |
BP 100785 Pa T 29.53 C | Z 0.48 raw 0.50 | AH en=1 act=1 tgt 0.45
err -0.03 dU -3.00 i 0.00 | M 1000 1000 1000 1000 us | GPS 0 sats=0
```

2. Message map

Message	ID	App use
HEARTBEAT	0	Connect / armed bit
ATTITUDE	30	3D view at 10 Hz
PARAM_*	20-23	Tuning gains
STATUSTEXT	253	Live Log (this document)
SD log list/dump	CLI / later	Separate flight-file download

3. Transport reminder

Item	Value
MAVLink	v2 clear frames, sys/comp 1/1
UART	PB6 TX / PB7 RX @ 115200
G20 (app on remote)	UDP listen 14551, send 127.0.0.1:14552
T12	Classic BT SPP after normal pair
ESP on PB6/PB7	Do not connect while using MAV

4. STATUSTEXT chunk protocol

STATUSTEXT.text is max 50 characters, so the FC splits the long line. severity = INFO (6).

text[0]	Meaning	App action
+	More chunks follow	Append text.substring(1) to a buffer
=	Final chunk	Append, flush one UI line, clear buffer

Dart reassembly sketch

```
String _logAcc = '';

void onStatusText(String text) {
  if (text.isEmpty) return;
  final mark = text[0];
  final body = text.length > 1 ? text.substring(1) : '';
  if (mark == '+') {
    _logAcc += body;
  } else if (mark == '=') {
    _logAcc += body;
    appendLiveLogLine(_logAcc.trim());
    _logAcc = '';
  }
}
```

5. Field meanings

Part in line	Meaning
[DMA] / [poll]	IMU sample path
sN	IMU sample sequence
DIS / ARM	Disarmed / armed
G gx gy gz dps	Body gyro (deg/s)
R P Y	Roll / pitch / yaw (degrees)
BP / T	Baro pressure (Pa) / temperature (C)
Z / raw	Altitude filtered / raw (m)
AH en act tgt err dU i	AltHold flags, target, error, corr (us)
M m1 m2 m3 m4 us	Motor PWM outputs (us)
GPS 0 sats=0	Placeholder until GPS is fitted

Keep GPS as 0 for now. Do not invent fix quality or satellite counts.

6. UI notes

- Scrolling Live Log list; keep about the last 300 lines
- Optional phone timestamp when a full line is flushed
- Pause / Clear / Share controls
- 3D view stays on ATTITUDE; Live Log shows the dense ASCII line
- Tuning still uses PARAM_*; do not parse gains from the log line
- Do not mix Live Log with SD-card flight log download

7. Acceptance (props off)

- Full line shows BP, Z, AH, motors, and GPS 0 sats=0
- Displayed line has no leftover + or = markers
- Update rate similar to Serial Monitor (~1-2 s)
- ESP is not sharing PB6/PB7

8. Chat one-liner

Live Log: chunked STATUSTEXT 253 (+ continue / = end) carries the full VCP status line including GPS 0 sats=0. Reassemble in Flutter. HB/ATTITUDE/PARAM unchanged. G20 UDP 14551 / 127.0.0.1:14552.